

FollowUpBoss Integration

1. Overview

1.1 Description

FollowUpBoss Integration connects **FollowUpBoss** (real estate CRM for agents and teams) with **Newo Platform**.

It enables:

- Creating or updating leads in the CRM with auto-deduplication via `POST /events`
- Looking up existing contacts by phone number to personalize conversations (lead stage, assigned agent)
- Checking appointment availability by querying existing appointments and computing free slots
- Scheduling appointments for showings, consultations, and meetings
- Cancelling existing appointments
- Automatically logging calls and conversation summary notes to contact records after the session ends

This integration is designed for **real estate agents and team administrators** who want to **capture leads and manage appointments through an AI receptionist** (voice or chat).

1.2 Use Cases

- When a caller provides their contact info as a new lead → Create lead via `POST /events` with source tracking
- When a caller is an existing contact → Look up their profile, display lead stage and assigned agent in the conversation
- When a caller asks about available times → Query existing appointments, compute free 30-min slots in 9-17 window, present options
- When a caller books a time → Create appointment in FUB with person ID and assigned agent
- When a caller cancels an appointment → Cancel via `DELETE /appointments/{id}`
- When a session ends → Automatically log the AI call and attach a conversation summary note to the contact

1.3 Supported Features

Feature	Supported	Notes
Lookup Existing Contact	✓	By phone number via <code>GET /people?phone=</code>
Create / Update Lead	✓	Via <code>POST /events</code> (auto-deduplicates by phone/email)
Check Availability	✓	Computes free slots from existing appointments (9-17 window)
Book Appointment	✓	<code>POST /appointments</code> with invitees (person + user)
Cancel Appointment	✓	<code>DELETE /appointments/{id}</code>
Post-Call Log	✓	<code>POST /calls</code> with outcome, duration, note (fires on session end)

Post-Call Note	✓	POST /notes with conversation summary (fires on session end)
Appointment Types Catalog	✓	Cached at setup via GET /appointmentTypes
Users / Agents Catalog	✓	Cached at setup via GET /users
Auto-Identify by Phone	✓	Fires on define_user_phone_number event
Reschedule Appointment	✗	Cancel + rebook as workaround (Phase 2)
Create Follow-Up Task	✗	Planned for Phase 2 (POST /tasks)

2. Requirements

Before installation:

- Active **FollowUpBoss** account (trial or paid)
- **FollowUpBoss API Key** (Admin → API)
- Optional **X-System** and **X-System-Key** if your team registered a system identity with FollowUpBoss

3. How to Setup

3.1 Step 1 — Obtain FollowUpBoss Credentials

1. Log in to your **FollowUpBoss** account
2. Navigate to **Admin → API** to get your **API Key**
3. Register your integration at <https://apps.followupboss.com/system-registration> to obtain:
 - **X-System** (system name, e.g., NewoAI)
 - **X-System-Key** (system-level API key)
4. Store all credentials securely

3.2 Step 2 — Connect in Platform

1. Open **Newo → Projects**
2. Import `followupboss.zip` to the project
3. Set the following attributes in **Builder / Attributes**:

Attribute	Required	Description
fub_api_key	✓	FollowUpBoss API key (Admin → API)
fub_x_system	✗	Optional registered system name (X-System header)
fub_x_system_key	✗	Optional registered system API key (X-System-Key header)
fub_base_url	✗	API base URL (default: https://api.followupboss.com/v1)
fub_default_event_source	✗	Lead source name (default: AI Receptionist)
fub_default_appointment_type_id	✗	Default appointment type (auto-populated from catalog)

fub_default_agent_user_id	✗	Default agent to assign (auto-populated from users catalog)
fub_setup_scenarios	✗	Auto-add / refresh FollowUpBoss canvas scenarios (default: True)
fub_enable_contact_lookup	✗	Enable fub_lookup_contact custom tool (default: True)
fub_enable_availability	✗	Enable fub_check_availability custom tool (default: True)
fub_enable_booking	✗	Enable fub_book_appointment custom tool (default: True)
fub_enable_cancellation	✗	Enable fub_cancel_appointment custom tool (default: True)
fub_enable_lead_capture	✗	Enable fub_create_lead custom tool (default: True)
fub_enable_post_call_updates	✗	Master switch for post-call CRM updates (default: True)
fub_enable_post_call_log	✗	Log call to contact timeline after session (default: True)
fub_enable_post_call_note	✗	Add conversation note after session (default: True)

4. Click **Save + Publish All**

5. On publish, the SetupFlow automatically:

- Creates / reuses HTTP connector (`fub_connector`)
- Registers custom tools through `_setupToolsSkill`
- Sends `refresh_static_data` to `CacheDataFlow` using the setup actor
- Refreshes appointment types and users from FollowUpBoss for dropdown-backed setup fields
- Refreshes canvas scenarios and intent types through `CanvasFlow` when `fub_setup_scenarios=True`

4. How to Use

4.1 How the Integration Works

The integration operates based on **Triggers and Actions** via the event-driven system.

- A **Trigger** is a system event that starts a flow
- An **Action** is the API operation performed in FollowUpBoss

Triggers

Trigger	Event	Description
Setup	<code>project_publish_finish</code>	Initializes integration on deploy
Canvas Setup	<code>gm_workflow_builder_canvas_built</code>	Adds custom intents/scenarios to canvas

Static Data Refresh	refresh_static_data	Refreshes appointment types and users into setup dropdowns
Auto-Identify Caller	define_user_phone_number	Looks up contact when phone is detected
Lookup Contact Tool	fub_lookup_contact	LLM-triggered contact lookup
Create Lead Tool	fub_create_lead	LLM-triggered lead creation
Check Availability	fub_check_availability	LLM-triggered availability check
Book Appointment	fub_book_appointment	LLM-triggered appointment creation
Cancel Appointment	fub_cancel_appointment	LLM-triggered cancellation
Post-Call Actions	custom_session_ended	Fires on session end → log call + add note

Available Actions

- **Search People** — GET /people?phone={phone}&limit=1
- **Create/Update Lead** — POST /events (auto-deduplicates)
- **Get Appointment Types** — GET /appointmentTypes
- **Get Users** — GET /users
- **Create Appointment** — POST /appointments with invitees
- **List Appointments** — GET /appointments?fromDate=...&toDate=... (for availability)
- **Cancel Appointment** — DELETE /appointments/{id}
- **Log Call** — POST /calls with personId, outcome, note
- **Add Note** — POST /notes with personId, subject, body

4.2 Architecture Diagrams

Overall Sequence

```
sequenceDiagram
    participant U as User
    participant A as ConvoAgent
    participant I as FollowUpBoss Integration
    participant API as FollowUpBoss API

    Note over I: Setup (on publish)
    A->>I: project_publish_finish
    I->>API: GET /appointmentTypes
    API-->>I: Appointment types
    I->>API: GET /users
    API-->>I: Users / Agents
    I->>I: Cache static data + register tools

    Note over U,API: Contact Identified
```

```

U->>A: (phone number detected)
A->>I: define_user_phone_number
I->>API: GET /people?phone={phone}
API-->>I: Contact data (or empty)
I->>A: ExistingClientDetails updated + urgent_message

```

Note over U,API: Availability Check

```

U->>A: "What times are available April 10?"
A->>I: get_available_slots {date, duration}
I->>API: GET /appointments?fromDate=...&toDate=...
API-->>I: Existing appointments
I->>I: Compute 30-min slots in 9-17 range, minus busy
I->>A: result_success {availability[]}
A->>U: "Here are available times: 9:00, 9:30, ..."

```

Note over U,API: Booking

```

U->>A: "Book 11 AM"
A->>I: book_slot {personId, appointmentTypeId, dateTime}
I->>API: POST /appointments
API-->>I: appointment_id
I->>A: result_success {booking_id}
A->>U: "Booking confirmed!"

```

Note over U,API: Cancellation

```

U->>A: "Cancel my appointment"
A->>I: convo_agent_cancel_booking {booking_id}
I->>API: DELETE /appointments/{id}
API-->>I: 204 No Content
I->>A: result_success {cancelled}
A->>U: "Your appointment has been cancelled"

```

Note over U,API: Session Ends

```

A->>I: custom_session_ended
I->>API: POST /calls {personId, outcome, note}
I->>API: POST /notes {personId, subject, body}

```

5. Project Structure

```

followupboss/
├── FollowUpBossIntegration/
│   ├── FollowUpBossIntegration.yaml
│   ├── libraries/
│   │   └── base64/                # encode/decode helpers (Basic Auth)
│   └── agents/
│       └── FollowUpBossAgent/
│           ├── agent.yaml
│           └── flows/
│               ├── SetupFlow/      # Attributes, connectors, tools, setup persona
│               ├── ApiFlow/        # Feature-routed HTTP adapter
│               └── CacheDataFlow/   # Static-data refresh for appointment types

```

```

and users
|
|   └─ AvailabilityFlow/  # fub_check_availability → GET /appointments →
compute free
|
|   └─ BookingFlow/      # fub_book_appointment → POST /appointments
|   └─ CancellationFlow/ # fub_cancel_appointment → DELETE
/appointments/{id}
|
|   └─ ExistingClientFlow/ # Auto-identify + manual lookup tool
|   └─ LeadFlow/          # fub_create_lead → POST /events
|   └─ PostCallFlow/      # custom_session_ended → log call + add note
|   └─ CanvasFlow/        # Canvas intents/scenarios
|   └─ MetadataFlow/      # Version info
└─ akb/
    └─ FollowUpBossAgent.yaml

```

6. Authentication

FollowUpBoss uses **two-layer authentication**:

Layer 1 — System Identification (optional)

When configured, requests include:

- `X-System` : Registered system name
- `X-System-Key` : System API key

Layer 2 — User Authentication (Basic Auth)

HTTP Basic Auth with API key as username, empty password:

```
Authorization: Basic base64(api_key:)
```

The integration includes a **custom base64 library** (`libraries/base64/encode.nsl`) because NSL Jinja2 doesn't expose Python's `base64` module.

7. Canvas Setup

On publish, `CanvasFlow` adds FollowUpBoss-specific intent types and scenarios to the canvas:

Intent Types

- `fub_appointment_booking_intent` — Schedule appointment
- `fub_lead_capture_intent` — New lead
- `fub_existing_contact_lookup_intent` — Look up existing contact
- `fub_property_inquiry_intent` — Property question
- `fub_cancellation_intent` — Cancellation request

Scenarios

- `fub_appointment_booking_scenario` — Booking flow with availability / booking code-phrases
- `fub_lead_capture_scenario` — Lead capture with `Let me save your information in FollowUpBoss now.`
- `fub_existing_contact_lookup_scenario` — Contact identification
- `fub_property_inquiry_scenario` — Property Q&A + lead capture fallback
- `fub_answering_questions_scenario` — General Q&A

- `fub_cancellation_scenario` — Cancellation with `I'm cancelling your booking now.`

Platform defaults are removed from canvas to avoid conflicts:

- `scenario_sales_appointment_via_calendar` (Google Calendar tool)
- `scenario_home_service_appointment_request`
- `scenario_common_schedule_appointment_agent`
- `intent_type_appointment_scheduling`, etc.

8. Testing

Unit / Integration Tests

```
poetry run pytest test/followupboss/ -v
```

Current source test count: **54 tests**.

Mock Server

A mock FollowUpBoss API server is available for local testing:

```
make mock-run PRODUCT=followupboss      # Starts server + ngrok tunnel
curl http://localhost:9090/_mock/routes  # List all mock routes
```

Routes defined in `mocks/followupboss/routes.yaml` .

E2E Scenarios

Live end-to-end tests verified on STG:

- `existing_contact_auto_identify` — lookup by phone → persona populated
- `appointment_booking_full_flow` — lookup → availability → book
- `booking_and_cancellation` — existing contact: book → cancel
- `booking_cancellation_new_customer` — new contact: availability → book → cancel
- `new_lead_capture` — new contact → `POST /events` via code-phrase
- `post_call_log_note` — session end → `POST /calls` + `POST /notes`

Scenario protocols in `test/followupboss/end_to_end/` .

9. Rate Limits (FollowUpBoss API)

Sliding 10-second window:

- **Global:** 250 req/10s (with X-System-Key), 125 without
- **POST /events:** unlimited
- **Notes:** 10 req/10s
- **PUT /people:** 25 req/10s

On HTTP 429, `Retry-After` header indicates wait time.

10. Known Limitations

- **Google Calendar hijack** — If customer has `project_attributes_setting_calendar_event_integration=google_integration` enabled, Google Calendar may intercept booking before FUB flow fires. SetupCanvasSkill deletes `scenario_sales_appointment_via_calendar` to prevent this.
- **Appointment reading** — FUB API only returns appointments created IN FUB (not synced from Google Calendar) and belonging to the requesting API user.
- **Session timeout** — Post-call flow fires on `custom_session_ended` , which is platform-driven (`project_attributes_setting_conversation_session_timeout_unfinished_chat` , default 300s).